

Simultaneous AUS/AUC Renormings

Audit of the proposed averaging argument, a corrected quantitative theorem, and the exact separable endpoint

Research note

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Abstract

We examine the proposed improvement of Lemma 4.3 in Dilworth–Kutzarova–Lancien–Randrianarivony. The proposed tail summation is a useful observation, but the smoothness estimate used in the packet has an unsupported dependence on the mixing parameter when the AUS exponent exceeds 2. For the predual of $N^* + \varepsilon M^*$, the generally valid power-type estimate supplied by duality is $\bar{\rho}(t) \lesssim \varepsilon^{-(p-1)} t^p$. The stronger $\varepsilon^{-1} t^p$ estimate used in the packet does not follow from the stated hypotheses when $p > 2$. Consequently, the claimed theorem for every $r > 2q$ is justified by that proof when $p \leq 2$, but not for arbitrary p .

We prove a corrected theorem valid for every reflexive Banach space: separate p -AUS and q -AUC renormings imply a common norm which is p -AUS and r -AUC for every $r > pq$. We also prove that the whole class of positive linear Asplund averages built from $N^* + \varepsilon_n M^*$ has an intrinsic pq -barrier.

Finally, the exact endpoint has a positive answer for separable reflexive spaces. This follows from Proposition 3.3 and Theorem 4.1(e) of Odell–Schlumprecht: the two separate norms give exact (q, p) -tree estimates (the lower and upper branches are allowed to be different), and hence an embedding into a reflexive space with an FDD satisfying exact 1- (q, p) estimates. Restricting the FDD norm gives a single p -AUS, q -AUC norm on the original space. I found no published resolution or counterexample for the arbitrary nonseparable case posed in the source paper.

1 Question and conclusions

For a normed Banach space X , write $\bar{\rho}_X$ and $\bar{\delta}_X$ for the moduli of asymptotic uniform smoothness and convexity. The question in Remark 4.4 of [6] is:

If a reflexive Banach space admits an equivalent norm with $\bar{\rho}(t) \leq Ct^p$ and, possibly under another equivalent norm, an estimate $\bar{\delta}(t) \geq ct^q$, must it admit one equivalent norm satisfying both estimates with the same exponents p and q ?

Note that $n_0 \leq \alpha t^{-q} + 1 \leq t^{-q}(\alpha + 1)$. Then, summing equations (4.2), (4.3) and (4.4) we obtain that for all $y \in B_E$ with $|y| \geq t$

$$|x + y| \geq |x| + \frac{D}{2n_0^3} t^q \geq 1 + \frac{D}{2(\alpha + 1)^3} t^{4q},$$

which concludes our proof. $\square$

Remark 4.4. *It follows from Corollary 2.4 that we could also build an equivalent norm which is AUC of power type q and AUS with a power type in $(1, p)$. We do not know if it is possible to preserve both power types together. To the best of our knowledge the similar question for uniformly smooth and convex norms is still open.*

Remark 4.5. *It is worth mentioning that one can use a nice Baire category argument to show that if a reflexive Banach space X admits an equivalent AUS norm and an equivalent AUC norm then it admits an equivalent norm which is simultaneously AUS and AUC. However, the very nature of Baire's Lemma makes it quite hopeless to get a quantitative result such as Lemma 4.3 so easily. Let us nevertheless describe this qualitative argument.*

Proof. We shall adopt here the presentation of the Asplund averaging technique, based on Baire's Lemma that can be found in [4] (p. 52-56). Let us denote by P the set of norms on X that are equivalent to the original norm $\|\cdot\|$ and let B be the closed unit ball of $\|\cdot\|$. For $N, M \in P$, we set

The outcome of the audit is summarized below. Here “general” includes nonseparable reflexive spaces.

Statement	Verdict	Reason
The submitted claim: common p -AUS and r -AUC for every $r > 2q$	Correct for $p \leq 2$; gap in general	The mixed norm has smoothness constant of order n^{p-1} , not n .
A corrected general averaging result	Proved here	A common p -AUS and r -AUC norm exists for every $r > pq$.
Exact preservation for separable reflexive spaces	Positive; implicit in the literature	Odell–Schlumprecht tree estimates and their exact FDD embedding theorem.
Exact preservation for arbitrary nonseparable reflexive spaces	No resolution found	The separable FDD theorem does not directly cover nonseparable spaces, and the averaging method has a quantitative barrier.

The correction is not cosmetic. If $p > 2$, the estimate asserted as equation (1) in the submitted packet does not follow from AUS/AUC duality with its stated dependence on n . The same scaling issue occurs in the displayed argument of Lemma 4.3 in [6]. This identifies a gap in that particular proof for large p ; it does not by itself disprove the theorem stated there.

2 Definitions and elementary facts

For $t > 0$, the asymptotic moduli of a norm $\|\cdot\|$ are

$$\begin{aligned} \bar{\rho}_X(t) &= \sup_{x \in S_X} \inf_{Y \in \text{cof}(X)} \sup_{y \in S_Y} (\|x + ty\| - 1), \\ \bar{\delta}_X(t) &= \inf_{x \in S_X} \sup_{Y \in \text{cof}(X)} \inf_{y \in S_Y} (\|x + ty\| - 1), \end{aligned}$$

where $\text{cof}(X)$ is the family of closed finite-codimensional subspaces of X . We say that the norm is p -AUS if $\bar{\rho}_X(t) \leq Ct^p$ for $0 < t \leq 1$, and q -AUC if $\bar{\delta}_X(t) \geq ct^q$ on that interval.

We use two standard facts.

- (i) If X is reflexive, a norm on X is p -AUS if and only if its dual norm is weak-star AUC of power type $p' = p/(p-1)$, up to universal changes of constants. More quantitatively, the two moduli are equivalent through Young duality; see [7, 6].
- (ii) AUS and AUC estimates pass to closed subspaces. Equivalent changes of norm alter constants, but not a fixed power exponent.

Throughout the quantitative argument we choose equivalent norms N and M satisfying

$$N(x) \leq M(x) \leq C_0 N(x), \quad \bar{\delta}_N(t) \geq at^q, \quad \bar{\rho}_M(t) \leq bt^p. \quad (1)$$

Then

$$C_0^{-1} N^*(x^*) \leq M^*(x^*) \leq N^*(x^*). \quad (2)$$

3 Audit of the submitted argument

The packet defines, for $n \geq 1$,

$$L_n^*(x^*) = N^*(x^*) + n^{-1} M^*(x^*)$$

and lets L_n be the predual norm. Its key smoothness assertion is

$$\bar{\rho}_{L_n}(t) \leq Knt^p. \quad (3)$$

The tail summation in the convexity half is sound once a suitable smoothness estimate is available. The problem is the exponent of n in (3).

3.1 The correct scaling

Let $p' = p/(p-1)$. Since M is p -AUS, M^* is weak-star AUC of power type p' . For $0 < \varepsilon \leq 1$, put

$$L_\varepsilon^* = N^* + \varepsilon M^*, \quad (4)$$

and denote its predual norm by L_ε .

Lemma 3.1 (Mixed-dual scaling). *There are constants $c, C > 0$, depending only on the data in (1), such that for all $0 < \varepsilon \leq 1$,*

$$\bar{\delta}_{L_\varepsilon^*}^*(s) \geq c\varepsilon s^{p'} \quad (0 < s \leq 1), \quad (5)$$

$$\bar{\rho}_{L_\varepsilon}(t) \leq C\varepsilon^{-(p-1)} t^p \quad (0 < t \leq 1). \quad (6)$$

Moreover,

$$(1 + \varepsilon)^{-1} N(x) \leq L_\varepsilon(x) \leq N(x). \quad (7)$$

Proof. The estimate (7) follows immediately from $N^* \leq L_\varepsilon^* \leq (1 + \varepsilon)N^*$, which in turn follows from (2).

We sketch the quantitative point in (5). Let $L_\varepsilon^*(x^*) = 1$. By (2), both $N^*(x^*)$ and $M^*(x^*)$ are uniformly controlled, and $M^*(x^*)$ is uniformly bounded away from zero. Use weak-star AUC of M^* to choose a weak-star closed finite-codimensional subspace on which the M^* -part gains at least a constant multiple of $s^{p'}$. Since X is reflexive, choose $u \in X$ which norms x^* for N^* , and intersect the preceding subspace with $\ker u$. On this intersection the N^* -part cannot decrease, while the M^* -part gains $c_0 s^{p'}$. Multiplication by the coefficient ε in (4) gives (5).

The Young conjugate of $s \mapsto c\varepsilon s^{p'}$ is, up to a constant, $t \mapsto \varepsilon^{-(p-1)}t^p$. Quantitative AUS/weak-star-AUC duality therefore gives (6). $\square$

Putting $\varepsilon = n^{-1}$ in Lemma 3.1 gives

$$\bar{\rho}_{L_n}(t) \leq Cn^{p-1}t^p, \quad (8)$$

whereas the argument in the packet requires Cnt^p . When $p \leq 2$, the weaker bound $n^{p-1} \leq n$ recovers the estimate used in the packet. When $p > 2$, the asserted dependence on n is not supplied by the cited duality argument. Thus the proof has a genuine gap; this observation alone does not show that the stronger estimate is false for every possible pair of norms.

3.2 What survives from the packet

If the averaging weights are $n^{-\theta}$, the smoothness sum is

$$\sum_{n=1}^{\infty} n^{-\theta} n^{p-1} = \sum_{n=1}^{\infty} n^{p-1-\theta},$$

so the correct condition is

$$\theta > p, \quad (9)$$

not merely $\theta > 2$. The convexity tail computed in the packet remains

$$t^q \sum_{n \gtrsim t^{-q}} n^{-\theta} \asymp t^{q\theta}.$$

Thus the same architecture works whenever one can choose $p < \theta < r/q$, namely whenever $r > pq$. In particular, the submitted $r > 2q$ statement is proved by that argument for $p \leq 2$, and more generally for an individual r satisfying $r > pq$.

4 A corrected general theorem

Theorem 4.1. *Let X be a reflexive Banach space. Suppose that X admits an equivalent p -AUS norm and an equivalent q -AUC norm, where $1 < p, q < \infty$. Then, for every $r > pq$, X admits an equivalent norm which is simultaneously p -AUS and r -AUC.*

Proof. Fix $r > pq$, and choose

$$p < \theta < r/q. \quad (10)$$

For $n \geq 1$, let L_n be the predual of $N^* + n^{-1}M^*$, and define

$$|x| = \sum_{n=1}^{\infty} n^{-\theta} L_n(x). \quad (11)$$

By (7), this is an equivalent norm, and there are constants $A, B > 0$ such that, uniformly in n ,

$$A|x| \leq L_n(x) \leq B|x|, \quad A|x| \leq N(x) \leq B|x|. \quad (12)$$

Smoothness. Let $|x| = 1$ and $0 < t \leq 1$. Choose m so large that

$$Bt \sum_{n>m} n^{-\theta} \leq t^p. \quad (13)$$

For each $1 \leq n \leq m$, Lemma 3.1, homogeneity, and the uniform equivalence (12) give a finite-codimensional subspace E_n such that

$$L_n(x+y) \leq L_n(x) + C_1 n^{p-1} t^p \quad (14)$$

whenever $y \in E_n$ and $|y| = t$. The constant C_1 is independent of n, m, x, t . Put $E = \bigcap_{n \leq m} E_n$. For $y \in E$ with $|y| = t$, use (14) on the head and the triangle inequality on the tail:

$$\begin{aligned} |x+y| &\leq \sum_{n \leq m} n^{-\theta} (L_n(x) + C_1 n^{p-1} t^p) + \sum_{n > m} n^{-\theta} (L_n(x) + L_n(y)) \\ &\leq 1 + C_1 t^p \sum_{n=1}^{\infty} n^{p-1-\theta} + Bt \sum_{n > m} n^{-\theta} \\ &\leq 1 + C_2 t^p. \end{aligned}$$

The series converges because $\theta > p$. Hence $|\cdot|$ is p -AUS.

Convexity. Let $|x| = 1$ and $0 < t \leq 1$. By the q -AUC estimate for N , uniform equivalence, and the standard rescaling of the AUC modulus, there are $\eta > 0$ and $E_0 \in \text{cof}(X)$ such that

$$N(x+y) \geq N(x)(1 + \eta t^q) \quad (15)$$

for every $y \in E_0$ with $|y| = t$. Choose $\alpha > 0$ large and an integer n_0 with

$$n_0 - 1 \leq \alpha t^{-q} \leq n_0. \quad (16)$$

For $n \geq n_0$, equations (7), (15), and $L_n(x) \leq N(x)$ give

$$\begin{aligned} L_n(x+y) - L_n(x) &\geq \frac{N(x+y)}{1+n^{-1}} - N(x) \\ &\geq N(x) \left(\frac{1+\eta t^q}{1+n^{-1}} - 1 \right) \geq dt^q, \end{aligned}$$

provided α is sufficiently large. Here $d > 0$ is independent of x, t, n , using $N(x) \geq A$.

For each of the finitely many indices $n < n_0$, choose a norming functional f_n for x in the norm L_n , and put

$$E = E_0 \cap \bigcap_{n < n_0} \ker f_n.$$

Then $L_n(x+y) \geq L_n(x)$ for $n < n_0$ and $y \in E$. Therefore, for $y \in E$ with $|y| = t$,

$$\begin{aligned} |x+y| &\geq |x| + dt^q \sum_{n \geq n_0} n^{-\theta} \\ &\geq 1 + d_1 t^q n_0^{1-\theta} \geq 1 + d_2 t^{q\theta}. \end{aligned}$$

Thus the norm is $q\theta$ -AUC. Since $q\theta < r$ and $t^{q\theta} \geq t^r$ for $0 < t \leq 1$, it is also r -AUC. $\square$

Remark 4.2. For $p \leq 2$, Theorem 4.1 is weaker than the packet's valid conclusion $r > 2q$ only when $p < 2$: in that range the packet's use of the coarser bound n imposes $\theta > 2$, whereas the sharp scaling allows the better threshold $r > pq$. At $p = 2$, the two thresholds agree.

5 A barrier for positive linear Asplund averages

The preceding proof can be formulated with arbitrary parameters. Let $\varepsilon_n \downarrow 0$, let $w_n > 0$, and let L_{ε_n} be the predual of $N^* + \varepsilon_n M^*$. Consider

$$|x| = \sum_{n=1}^{\infty} w_n L_{\varepsilon_n}(x). \quad (17)$$

The termwise smoothness estimate from Lemma 3.1 suggests the necessary summability condition

$$\sum_{n=1}^{\infty} w_n \varepsilon_n^{-(p-1)} < \infty. \quad (18)$$

This condition forces a quantitative loss in the convexity tail.

Proposition 5.1 (Averaging barrier). *Assume (18), and define*

$$W(s) = \sum_{\varepsilon_n \leq s} w_n.$$

Then

$$W(s) = o(s^{p-1}) \quad (s \downarrow 0). \quad (19)$$

Consequently, the AUC gain furnished by the tail mechanism at scale $s = t^q$ is

$$t^q W(t^q) = o(t^{pq}). \quad (20)$$

In particular, no argument based solely on a positive linear average of the mixed norms and on summing their individual tail gains can yield a uniform lower bound of order t^{pq} , let alone t^q .

Proof. For $\varepsilon_n \leq s$, one has $w_n \leq s^{p-1} w_n \varepsilon_n^{-(p-1)}$. Hence

$$\frac{W(s)}{s^{p-1}} \leq \sum_{\varepsilon_n \leq s} w_n \varepsilon_n^{-(p-1)}.$$

The right-hand side tends to zero as $s \downarrow 0$, because it is the tail over those n for which ε_n is small of the convergent series in (18). This proves (19); setting $s = t^q$ gives (20). $\square$

Proposition 5.1 is a limitation of this particular method, not a counterexample to exact simultaneous renorming. It explains why merely optimizing the weights in the submitted construction cannot settle the endpoint question.

6 The exact endpoint for separable reflexive spaces

The exact separable case is already contained, implicitly, in the tree/FDD machinery of Odell and Schlumprecht [10]. We spell out the deduction because it is very close to the question in Remark 4.4 but does not appear to have been stated there in this form.

6.1 The order of the exponents

Lemma 6.1. *If an infinite-dimensional reflexive Banach space admits a p -AUS renorming and a q -AUC renorming, then $p \leq q$.*

Proof. Choose a normalized weakly null basic sequence. A q -AUC norm yields, after passage to a subsequence, a lower ℓ_q estimate; a p -AUS norm yields, after a further passage to a subsequence, an upper ℓ_p estimate. These are the sequential special cases of [10, Proposition 3.3]. Equivalent norms only change the constants. Thus for some $c, C > 0$, the final subsequence (x_i) satisfies

$$c \left(\sum |a_i|^q \right)^{1/q} \leq \left\| \sum a_i x_i \right\| \leq C \left(\sum |a_i|^p \right)^{1/p}.$$

Taking $a_1 = \dots = a_n = 1$ gives $cn^{1/q} \leq Cn^{1/p}$ for every n , hence $p \leq q$. $\square$

6.2 Tree estimates and exact FDD estimates

Odell–Schlumprecht say that X satisfies C -(P, Q)-tree estimates if every normalized weakly null tree has one branch with a lower ℓ_P estimate and (possibly a different) branch with an upper ℓ_Q estimate. The permission to use two different branches is the key point.

Their Proposition 3.3 gives:

$$\begin{aligned} q\text{-AUC} &\implies (q, 1)\text{-tree estimates,} \\ p\text{-AUS} &\implies (\infty, p)\text{-tree estimates.} \end{aligned}$$

After fixing either original norm as a reference, normalize the nodes in the other norm before applying Proposition 3.3 and then use equivalence of the two norms to return to the reference normalization. Thus the two separate renormings give (q, p) -tree estimates in one fixed reference norm. Lemma 6.1 supplies the required order $p \leq q$.

The exact form of [10, Theorem 4.1(e)] says that a separable reflexive space with (P, Q) -tree estimates embeds into a reflexive space having an FDD with exact 1-(P, Q) estimates. Applying it with $(P, Q) = (q, p)$ gives a space Z with an FDD (E_i) such that every finite block sequence (z_i) satisfies

$$\left(\sum_i \|z_i\|^q \right)^{1/q} \leq \left\| \sum_i z_i \right\| \leq \left(\sum_i \|z_i\|^p \right)^{1/p}. \quad (21)$$

Lemma 6.2. *If a Banach space Z has an FDD satisfying (21), then its norm is p -AUS and q -AUC.*

Proof. Fix $x \in S_Z$ and $0 < t \leq 1$. Given $\eta > 0$, choose a finitely supported vector u with $\|x - u\| < \eta$, and let Y be the tail of the FDD after the support of u . For $y \in S_Y$, the two vectors u and ty form a block sequence, so

$$\begin{aligned} \|x + ty\| &\leq \eta + (\|u\|^p + t^p)^{1/p}, \\ \|x + ty\| &\geq (\|u\|^q + t^q)^{1/q} - \eta. \end{aligned}$$

In the first inequality choose η to be a sufficiently small multiple of t^p ; in the second choose it to be a sufficiently small multiple of t^q . Since $1 - \eta \leq \|u\| \leq 1 + \eta$, elementary estimates for $(a^s + t^s)^{1/s}$ yield constants $C, c > 0$, independent of x, t , such that

$$\sup_{y \in S_Y} \|x + ty\| \leq 1 + Ct^p, \quad \inf_{y \in S_Y} \|x + ty\| \geq 1 + ct^q.$$

This is precisely p -AUS and q -AUC. $\square$

Theorem 6.3 (Exact separable preservation). *Let X be a separable reflexive Banach space. If X admits separately an equivalent p -AUS norm and an equivalent q -AUC norm, then X admits one equivalent norm which is simultaneously p -AUS and q -AUC.*

Proof. The finite-dimensional case is immediate. Assume that X is infinite-dimensional. As above, X satisfies (q, p) -tree estimates. By [10, Theorem 4.1(e)], it embeds into a reflexive Z with an FDD satisfying exact 1 -(q, p) estimates. Lemma 6.2 shows that the norm of Z is p -AUS and q -AUC. Restricting this norm to the embedded copy of X preserves both estimates. $\square$

Remark 6.4. The paper [10] explicitly works with separable Banach spaces. Thus Theorem 6.3 does not by itself answer the nonseparable question in [6]. It does show that any obstruction to the general endpoint must be genuinely nonseparable, rather than a phenomenon already visible on one separable subspace.

7 Literature search and status of the nonseparable problem

The literature search used the exact wording of Remark 4.4, variants of “preserve both power types,” citation searches around [6], and recent surveys and renorming papers. The main findings are:

- (1) The exact separable result follows from [10, Proposition 3.3 and Theorem 4.1(e)], as proved in Theorem 6.3. The crucial lines are the exact tree-to-FDD exponent transfer and the fact that lower and upper branches need not coincide.
- (2) Lancien’s 2017 lecture notes still list preservation of both exponents as a question and emphasize that the fixed-power conditions are not suitable for the elementary Baire-category proof used for qualitative simultaneous AUS/AUC renorming [9].
- (3) Recent work proves strong one-sided, separably determined renorming theorems for power-type AUS properties; see, for example, Causey–Fovelle–Lancien [4]. These results do not, as stated, produce one global norm simultaneously realizing a prescribed AUS exponent and a prescribed AUC exponent.
- (4) A 2024 paper of Baudier–Lancien describes the qualitative simultaneous AUS/AUC renorming theorem as the current route to property (β) , without providing an exact prescribed-power theorem [2].
- (5) The 2025 monograph of Baudier–Lancien [3] provides a current large-scale account of asymptotic renorming and the qualitative simultaneous AUS/AUC problem. I did not locate in it, or in papers citing [6], a published theorem or counterexample settling the exact endpoint for arbitrary nonseparable reflexive spaces.

The last statement is a report of the search, not a logical proof that no resolution exists. The safe current conclusion is therefore:

The exact endpoint is settled positively for separable reflexive spaces. For the arbitrary nonseparable setting of Remark 4.4, no published full answer or counterexample was found in this search. The positive linear Asplund averaging method cannot reach the endpoint, and its correct general output is Theorem 4.1.

8 Further directions

The analysis isolates two plausible routes beyond the present result.

First, one could seek a nonseparable analogue of the exact tree-to-FDD coordinatization theorem. The obstacle is that asymptotic moduli in a nonseparable space are naturally controlled by weakly null nets and finite-codimensional subspaces, while the Odell–Schlumprecht construction is sequential and explicitly separable.

Second, a successful averaging construction must be nonlinear in a substantial way. Proposition 5.1 rules out obtaining the endpoint by merely changing the positive weights or the sequence (ε_n) in a linear sum of the predual norms of $N^* + \varepsilon_n M^*$. Supremum-type, infimal-convolution, or scale-adaptive constructions would have to avoid both the summability condition (18) and the loss of AUC caused by a fixed mixing parameter.

Acknowledgment of scope

This note gives a complete audit of the submitted proof, a rigorous corrected quantitative theorem for arbitrary reflexive spaces, a method barrier, and a complete positive solution of the exact problem in the separable case. It does not claim to settle the remaining arbitrary nonseparable endpoint.

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